

Leetcode Question Pattern

Top k Form

Sorted Structure

Introduction:

$$\text{nums1} = \{9, 5, 7\}$$

$$\text{nums2} = \{8, 1, 6\}$$

$$\begin{matrix} 17 & 10 & 15 \\ (9, 8) & (9, 1) & (9, 6) \end{matrix}$$

$$\begin{matrix} 13 & 6 & 11 \\ (5, 8) & (5, 1) & (5, 6) \end{matrix}$$

$$\begin{matrix} 15 & 8 & 13 \\ (7, 8) & (7, 1) & (7, 6) \end{matrix}$$

you have to
make pair
from num1
& num2

$$\begin{matrix} \text{num1} & \text{num2} \\ \downarrow & \downarrow \\ (& , &) \end{matrix}$$

Now you to find

Top-4 pair sum:

$$\Rightarrow (9, 8), (9, 6)$$

$$(7, 8), (5, 8)$$

```

int m = num1.length();
int n = num2.length();

for(i=0; i<=m; i++){
    for(j=0; j<=n; j++){
        pair(num1[i], num2[j]);
        sum = num1[i] + num2[j];
    }
}
sort();
}

```

$O(n^2)$
 $\downarrow$
 TLE
 ✗

- (•) Generate every possible pair sum
- (•) Sorted everything
- (•) we picked top-k sum

- what if, I have to sort it:

nums1 = {9, 5, 7}

nums2 = {8, 1, 6}

¹⁷
(9, 8), ¹⁵
(9, 6), ¹⁰
(9, 1) → 17 15 10

¹⁵
(7, 8), ¹³
(7, 6), ⁸
(7, 1) → 15 13 8

¹³
(5, 8), ¹¹
(5, 6), ⁶
(5, 1) → 13 11 6
}
Sorted order

Important

- Top k
 - Smallest k
 - Largest k
- Once think about on
- HEAP

$$\text{nums1} = \{ \overset{0}{1}, \overset{1}{7}, \overset{2}{11} \}$$

$$k = 3$$

k pairs with
smallest sum.

$$\text{nums2} = \{ \overset{0}{2}, \overset{1}{4}, \overset{2}{6} \}$$

$$\text{smallest} : 1 + 2 = 3 \\ (0, 0)$$

$$\left. \begin{array}{l} \text{nums1} = \{ \overset{0}{1}, \overset{1}{7}, \overset{2}{11} \} \\ \text{nums2} = \{ \overset{0}{2}, \overset{1}{4}, \overset{2}{6} \} \end{array} \right\} \begin{array}{l} \text{which one} \\ \text{I Take} \end{array}$$

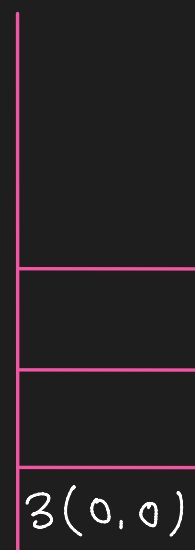
So here we maintain a heap:

$$\begin{array}{cc} i & j \\ 3(0, 0) \end{array}$$

$$i = 0$$

$$(i, j+1) \rightarrow (1, 4) = 5$$

$$(i+1, j) \rightarrow (7, 2) = 9$$



HEAP

$$\text{sum} = 5$$

$$i = 0$$

$$j = 1$$

similarly:

$$\begin{aligned} & \swarrow (0, 2) \quad (i, j+1) \rightarrow (1, 6) \\ & (i, j+1) \quad (i+1, j) \rightarrow (7, 4) \\ & \quad (1, 1) \end{aligned}$$

5(0, 1)
9(1, 0)

HEAP

$$\text{result} = \{ \overset{3}{\uparrow} (1, 2), \overset{5}{\uparrow} (1, 4), \overset{7}{\uparrow} (1, 6) \} \quad \swarrow$$

$$\text{sum} = 7$$

$$i = 0$$

$$j = 2$$

7(0, 2)
9(1, 0)
11(1, 1)

★ How to identify

- k^{th} smallest
- k^{th} largest
- Merge k sorted list/arrays etc.
- sorted rows in 2D matrix $\rightarrow$ find top k
- sorted column in 2D matrix $\rightarrow$ find top k
- pair sum $\rightarrow$ find top k
- sub-array sums $\rightarrow$ find top k
- Top k
- Best k